SUSTAINABILITY ASSESSMENT CONCENTRATION

[COURSE CODE]

Circularity and Life Cycle Assessment

See the whole life of a product, then redesign it.

Measure the life cycle impact, carbon footprint and circularity of what engineers build.

WHAT YOU WILL STUDY

- Circular economy principles and business models
- Measuring circularity and material flows
- Life cycle assessment (LCA): goal, scope and inventory analysis
- Impact assessment and interpretation
- Product carbon footprints and environmental product declarations
- LCA software and databases

YOU WILL BE ABLE TO

1. Carry out a comparative LCA that conforms to ISO standards
2. Calculate a product carbon footprint
3. Measure and improve the circularity of a product or system

ABBREVIATIONS

ISO International Organization for Standardization

LCA Life cycle assessment

AUDIENCE

Civil and Environmental Engineering

Materials Science and Engineering

Mechanical Engineering

Chemical Engineering

Industrial and Systems Engineering

Architectural Engineering

STANDARDS COVERED

ISO 14040 and ISO 14044
Life cycle assessment

ISO 14067
Carbon footprint of products

ISO 59000 series
Circular economy

PROFESSIONAL PATHWAY

Builds the skills of an LCA practitioner and carbon footprint analyst.

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